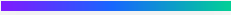


OpenPipeline, Segments, and IAM Policies

SE Training – January 2025

Training Recorded

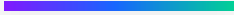


Warning This training is being recorded and will be available for future on-demand consumption. Please maintain professionalism at all times. Thank you.

Agenda

- Goal: To introduce the paradigm shift from classic Management Zones to data permissions and filtering on Grail, we'll configure OpenPipeline, Segments, and IAM Policies to provide role-based access and analysis controls
- Segments 101
- Paradigm Shift – Management Zones vs Segments + IAM
 - Common Scenarios and Solutions
- Hands On: Segments
 - Platform Engineering, Internal Development Platform, Self-Service Onboarding
 - Analyzing Data without Segments
 - Basic Segments
 - Kubernetes Cluster and Namespace
 - OpenPipeline
 - Complex Segments
 - Project Team
- Segments Roadmap
 - Official Roadmap
 - Internal RFE (When is XYZ coming?)

Segments 101



Why Segments?

Settings	
Monitoring	▼
Setup and overview	
Cloud Automation	▼
Setup and configuration	
Processes and containers	▼
Monitoring, detection and naming	
Web and mobile monitoring	▼
Real user and synthetic monitoring	
Cloud and virtualization	▼
Connect cloud and virtualization types	
Server-side service monitoring	▼
Manage and customize service monitoring	
Service Detection	▼
Define rules for services and spans	
Log Monitoring	▼
Set up management of logs	
Anomaly detection	▼
Configure detection sensitivity	
Alerting	▼
Configure alerting settings	
Dashboards	▼
Configure dashboard settings	
Metrics	

Management zones settings

Management zones enable defining fine grained access rights to parts of an environment. A Management zone consists of a set of entities like applications, hosts, process groups, or services.

[More...](#)

Your user does not have the necessary write permissions.

Filter items...

Summary	Details
FinOps – Host aks-default-77091117-vmss00001Y TEST	▼
mz-az-af	▼
mz-az-af-abssahara	▼
mz-az-agcs	▼
mz-az-agcs-cl	▼
mz-az-agcs-fa	▼
mz-az-agcs-ipp	▼
mz-az-agcs-middleware	▼

Page Unresponsive

You can wait for it to become responsive or exit the page.

Management zones settings - Environment - Settings - Allianz ASI...

Wait

Exit page

prod

Allianz ASIC

Filter

Select management zone

All

FinOps - Host aks-default-77091117-vmss00001...

mz-az-af

mz-az-af-abssahara

mz-az-agcs

mz-az-agcs-cl

mz-az-agcs-fa

mz-az-agcs-ipp

mz-az-agcs-middleware

mz-az-agcs-ms

mz-az-agcs-pa

mz-az-agcs-ra

mz-az-agcs-sd

mz-az-agcs-uw

mz-az-agcs-wp

mz-az-at

mz-az-at-appdev

mz-az-at-appdev_playground

mz-az-at-bmpproduct

mz-az-au

mz-az-au-aaldevops

mz-az-au-prod

mz-az-ay

mz-az-ay-digitalplatform

mz-az-azp

mz-az-azp-AZGACanadaIT

mz-az-azp-development

mz-az-bcm_de

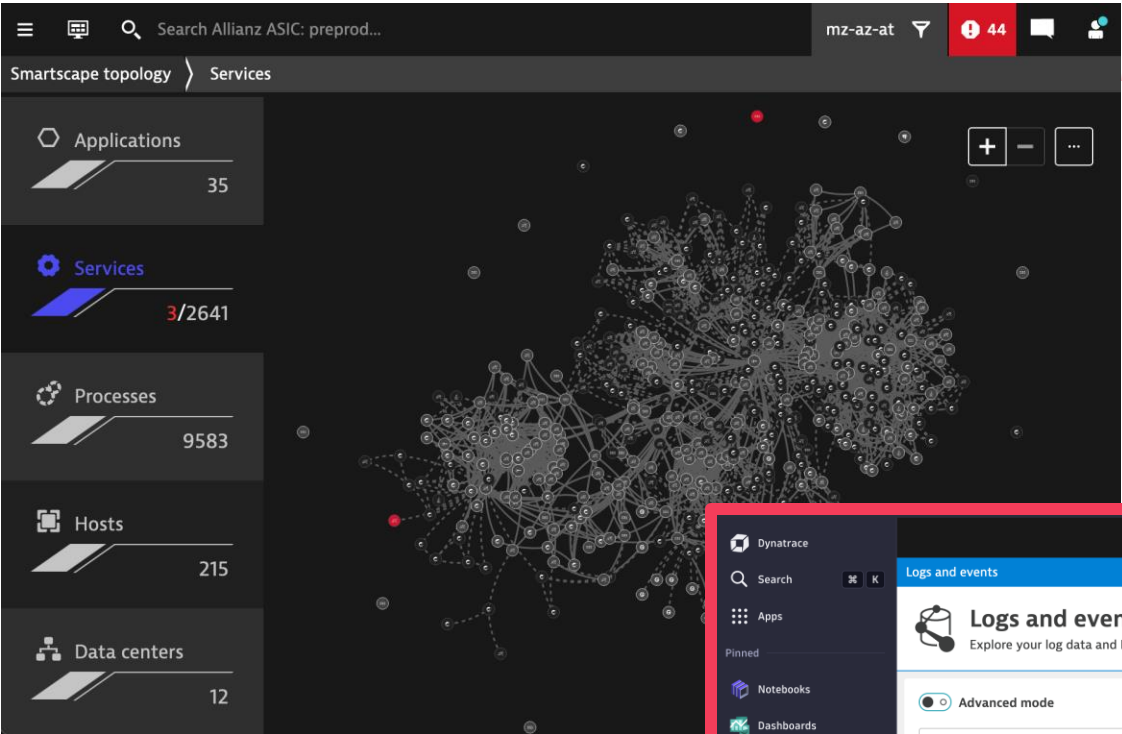
mz-az-bcm_de-dbitkvi5co

mz-az-bcm_de-global

mz-az-bg

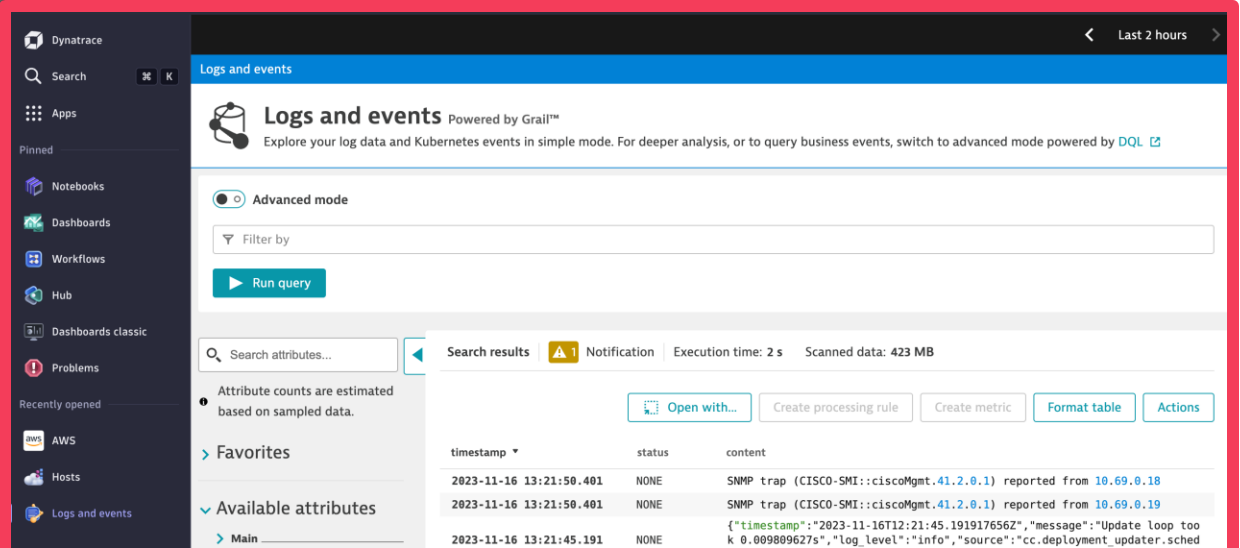
mz-az-bg-AZBG_DynaTrace

Why Segments?



Experience the full power of Grail and the Dynatrace Query Language

Where MZ?



Segments: Preconfigured Dynamic Filtering

Kubernetes

Explorer

Anomaly detectors

Cluster: CL-Prod20

Filter by:

+ Add filter

Clusters

Nodes

Namespaces

Workloads

Deployments

DaemonSets

StatefulSets

ReplicaSets

Replication controllers

CronJobs

Jobs

Pods

Services

Containers

Clusters

12 records

Davis® AI

Health status

Clusters

1 / 16

Nodes

4 / 456

Namespaces

137

Workloads

149 / 5.9k

Health

Utilization

Metadata

12 columns hidden

Cluster	Problems	Nodes	Namespaces	Workloads	Pods	CPU Usage	CPU Requests	CPU
code-matrix-cluster	2	12	8	789	1.2k			
quantum-compute-pod	1	1 / 2	5	12	78			
binary-nexus	-	30	12	13 / 568	234			
devops-grid	-	1	5	456	48			
cyberspace-core	-	2 / 500	23	1.3k	89			
robo-mesh-cluster	-	4	3	8 / 90	5			
nano-dene-forge	-	12	1	4 / 45	67			
crypto-mesh-hub	-	1 / 45	24	456	124			
data-pipeline-grid	-	5	4	545	78			
quantum-logic-nest	-	1	3	3	3			
technosphere-cluster	-	4	3	3 / 90	5			

1 click to block out all the noise

Segments Implementation

Trigger

Trigger

We offer two predefined variants:

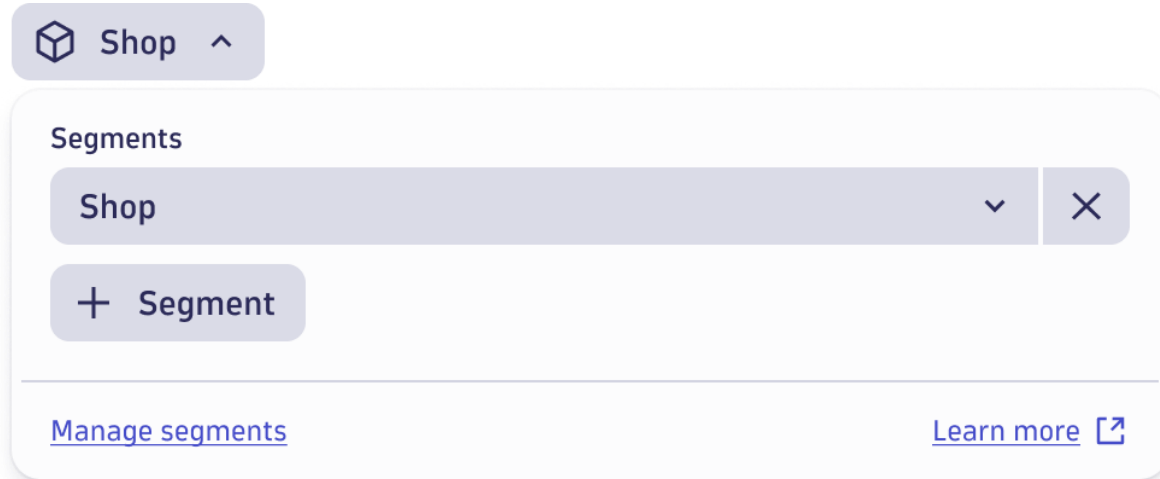
- Default, where we show the currently applied segment if one segment is selected. Multiple segments are grouped to make the width more predictable.
- Compact, where we only show the amount of selected segments.

Additionally, the trigger is customizable, as for all our overlays.

	No segment	Simple segment	Segment with variable (1 value)	Segment with variable (2+ values)	2+ segments
Default trigger	 ▾	 Shop ▾	 Region: eu-ireland ▾	 Region (2) ▾	 2 segments ▾
Compact trigger	 ▾	 1 ▾	 1 ▾	 1 ▾	 2 ▾



Simple Segment



Show me entities, logs, events, etc. of my application

- Equivalent to single MZ

Dynamic Segment

Region: eu-ireland ^

Segments

Region v eu-ireland, eu-austria ^ X

+ Segment

[Manage segments](#)

Search

- ☒ eu-ireland
- ☒ eu-austria
- ☐ eu-france
- ☐ eu-germany
- ☐ eu-italy
- ☐ eu-spain
- ☐ us-east
- ☐ us-west

Show me data of “eu-ireland” and “eu-austria” Regions

- Deduplication of MZs
- Dynamic conditions using variables
- Multi-select of Dynamic Segment values will result in OR logic when filtering
 - Filter region == “eu-Ireland” OR region == “eu-austria”

Multiple Segments

 2 segments ^

Segments

Region

eu-ireland, eu-austria

×

App-env

production

×

+ Segment

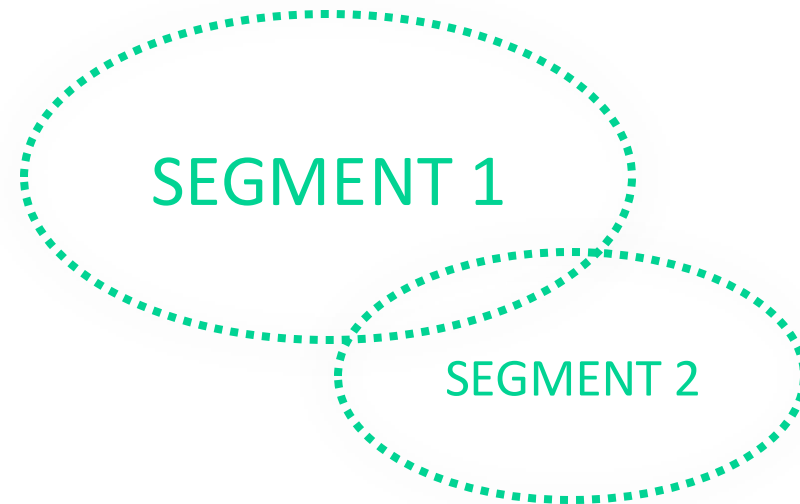
Clear all

[Manage segments](#)

[Learn more](#) 

Show me data of my application from a specific infrastructure

- Intersection of multiple segments
(eu-ireland OR eu-austria) AND production



How it all works

2 segments ^

Segments

Region	▼	eu-ireland, eu-austria	▼	×
App-env	▼	production	▼	×

+ Segment Clear all

[Manage segments](#) [Learn more](#)



1) Segment IDs and values are passed to query API

FiRmGMvWzPY(eu-ireland, eu-austria), HCkccp8aAIU(production)

2) Grail filters consumer queries with conditions defined in segments

```
fetch logs
| filter
  // permission-based filters
  ... AND
  // Segment-based filters
  (
    (condition_a1 OR condition_a2 OR ...) AND
    (condition_b1 OR condition_b2 OR ...)
  )
  // consumer query continues
| summarize ...
```

Transparent
filter conditions

Building a Segment

< All segments

This segment has unsaved changes.

Discard changes

Save

ACE DT

(Full-stack) Observability data for ACE DT team.

Owner

R Roman Windischhofer

Visibility

Anyone in the environment

2 Variables for dynamic segments

\$namespace.name

dynatrace

\$namespace.id

CLOUD_APPLICATION_NAM...

Variables

Create variables to apply in your Segment data filters

Segment data

Include all data that should be accessible when applying the Segment

Logs

3 Include data directly

k8s.namespace.name = \$namespace.name

Preview

Metrics

k8s.namespace.name = \$namespace.name

Preview

Hosts

4 Include data of entities

tags = team:ACE

Preview

Include

Problems

Vulnerabilities

Metrics

Logs

Events

BizEvents

Spans

Services

5 Include data of related entities

runs_on = \$this.hosts

Preview

Include

Problems

Vulnerabilities

Metrics

Logs

Events

BizEvents

Spans

+ All types

+ Metrics

+ Logs

+ Events

+ BizEvents

+ Spans

+ Entities

1 Visible to anyone or unlisted

20 include blocks per segment

1 filter for data of entities (logs, events, spans)

No limit for problems, vulnerabilities, metrics

1 topology traversal step only

14

Segments and in-app filters

24 Nodes3 / 3 Namespaces12 Workloads3.5k / 12.7k

CapabilitiesTopology

Node	Problems	Pods	Workloads	Services	Usage	Requests	alertmanager	apiserver	key	key
full-observability...	-	123	3	45			main	main	✓	-
full-observability...	-	12	3	345			longvaluetextexample...	✓	-	-
GKE CP KLU	-	3	12	3			main	main	✓	-
OCP 3.11	-	12	456	1			-	longvaluetextexample...	✓	-
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
ocp4-9-in-cluster	-	4	89	3			longvaluetext	longvaluetextexample...	-	-
ocp4-9-kuberne...	-	4	45	54			main	longvaluetext	-	-
rauter-cri-test	-	45	456	34			✓	✓	longvaluetextexample...	main
trauter-e2e	-	5	545	1,234			-	-	✓	longvaluetext e
full-observability...	-	123	3	45			main	main	✓	-
full-observability...	-	12	3	345			longvaluetextexample...	✓	-	-
GKE CP KLU	-	3	12	3			main	main	✓	-
OCP 3.11	-	12	456	1			-	longvaluetextexample...	✓	-
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
ocp4-9-in-cluster	-	4	89	3			longvaluetext	longvaluetextexample...	-	-
ocp4-9-kuberne...	-	4	45	54			main	longvaluetext	-	-
rauter-cri-test	-	45	456	34			✓	✓	longvaluetextexample...	main
trauter-e2e	-	5	545	1,234			-	-	✓	longvaluetext e
ocp4-12	-	467	24	12			-	-	longvaluetextexample...	-
trauter-e2e	-	1	3	3			-	✓	longvaluetextexample...	-

20 rows per page

1 Segments selector to set context

2 Additional filter for further problem analysis

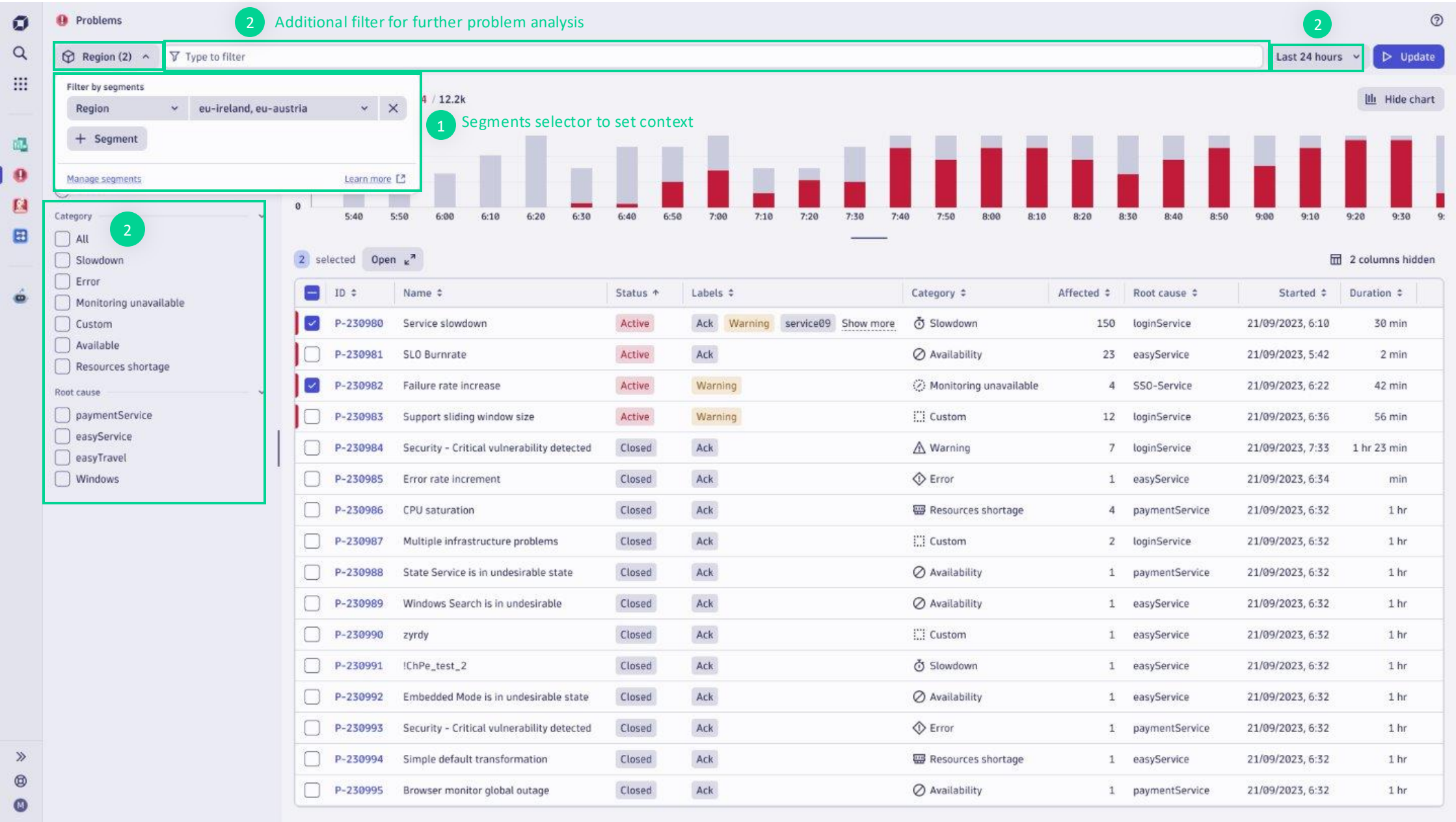
Type to filter

Update

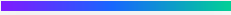
12 columns hidden

Page 1 of 2





Management Zones to Segments and IAM



Common Scenarios and Solutions

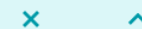
Application with Host Group

- Application deployed on dedicated infrastructure
- Monitored with OneAgent using Host Group configuration
 - Host group identifier based on application name, infrastructure layer, and environment
- Application team should only have access to data in Dynatrace related to their monitored application

easyTravel AWS



easyTravel AWS Linux



Management zone name

easyTravel AWS Linux

Be careful when renaming - if there are policies that are referencing this Management zone, they will need to be adapted to the new name!

Description

Add a description

Rules

Add a new rule

Enabled

Summary



Browser monitors (Synthetic monitor) on **Browser monitors** where Browser monitor tags tag key equals 'easytravel-aws-linux'



HTTP monitors (Synthetic monitor) on **HTTP monitors** where HTTP monitor tags tag key equals 'easytravel-aws-linux'



Process groups on Hosts where Host group name equals 'aws-easytravel-lnx'



Web applications (Application Entity) on **Web applications** where Web application name equals 'easyTravel Linux'

Display entities



Process groups on Hosts where Host group name equals 'aws-easytravel-lnx'

Rule type

Monitored entity



Rule applies to

Process groups



Conditions

Add condition

Include all process groups, hosts, and services that belong to our host group

Summary

Host group name equals 'aws-easytravel-lnx'



Apply to underlying hosts of matching process groups



Apply to all services provided by the process groups



View group easyTravel AWS Linux Group

View and edit group permissions.

Details

Edit

Group name
easyTravel AWS Linux Group

Group description

Group type
LOCAL

Permissions

Permission	Permission Type	Scope	Actio...
View environment	ROLE	Environment (hfn13693:easyTravel AWS Linux, tpc-demo)	⋮
Manage monitoring settings	ROLE	Environment (hfn13693:easyTravel AWS Linux,	⋮

10 rows per page

Page 1 of 1

Permission name

Manage monitoring settings

Permission description

Manage monitoring settings

Permission type

role

Scope

Environments

Select environments

Management zones

easyTravel AWS Linux

Summary

Role is scoped for 1 management zone: hfn13693:easyTravel AWS Linux

Classic role-based access to only view and/or manage entities/data within management zone

Delete group

Edit

+ Permission



Smartscape topology > Hosts > aws-et-ubuntu-01 Full Stack

Show problems ▾

Filter classic views, such
as Smartscape, on
management zone

Applications

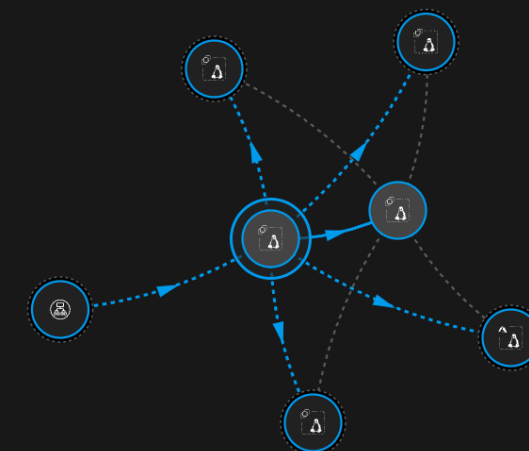
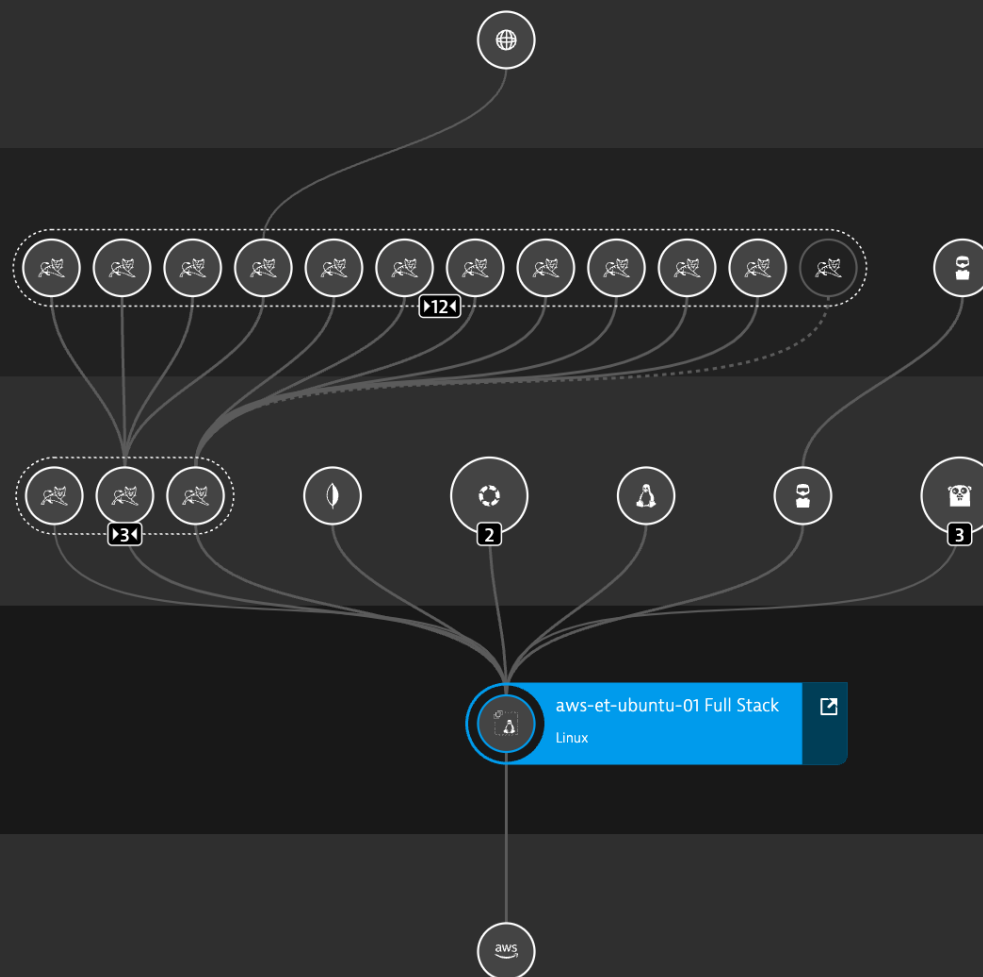
Services

Processes

Hosts

2

Data centers



View group easyTravel AWS Linux Group

View and edit group permissions.

Details

Group name

easyTravel AWS Linux Group

Group description

Group type

LOCAL

Edit

Permissions

Permission	Permission Type
View environment	ROLE
Manage monitoring settings	ROLE
AppEngine - Notebooks - Restricted User	POLICY
Data Access - AWS easyTravel Linux	POLICY
AppEngine - All Apps	POLICY

10 rows per page

Grail permissions based on IAM Policy statements

Policy name*

Data Access - AWS easyTravel Linux

Policy description

Read access to data in Grail for AWS easyTravel Linux

Policy statement*

1

2

4

5

6

ALLOW storage:buckets:read;

ALLOW storage:system:read;

ALLOW storage:entities:read WHERE storage:dt.security_context IN ('aws-easytravel-lnx', 'easyTravel AWS Linux');

ALLOW storage:events:read WHERE storage:dt.host_group.id = 'aws-easytravel-lnx';

ALLOW storage:logs:read WHERE storage:dt.host_group.id = 'aws-easytravel-lnx';

ALLOW storage:metrics:read WHERE storage:dt.host_group.id = 'aws-easytravel-lnx';

ALLOW storage:bizevents:read WHERE

Learn how to manage policy statements

Edit

Delete group

Permission

Actions



IAM Policy Statements – Storage (Grail)

- <https://docs.dynatrace.com/docs/manage/identity-access-management/permission-management/manage-user-permissions-policies/advanced/iam-policystatements#storage>

storage:entities:read

Grants permission to read records from entities

conditions:

- `storage:entity.type` - The type of the entity.
operators: `=`, `IN`, `startsWith`
- `storage:dt.security_context` - Custom field for security context.
operators: `=`, `IN`, `startsWith`

Read entities based on
dt.security_context value only until
entity model actually in Grail

storage:logs:read

Grants permission to read records from the logs-table

conditions:

- `storage:bucket-name` - This condition reduces the effect of the record-level permission to a defined list of buckets.
operators: `=`, `!=`, `IN`, `NOT IN`, `startsWith`, `NOT startsWith`
- `storage:k8s.namespace.name` - The name of the namespace that the pod is running in.
operators: `=`, `IN`, `startsWith`
- `storage:k8s.cluster.name` - The name of the cluster that the pod is running in.
operators: `=`, `IN`, `startsWith`
- `storage:host.name` - Name of the host.
operators: `=`, `IN`, `startsWith`
- `storage:dt.host_group.id` - Id of the host group.
operators: `=`, `IN`, `startsWith`
- `storage:log.source` - The location where the log comes from.
operators: `=`, `IN`, `startsWith`
- `storage:dt.security_context` - Custom field for security context.
operators: `=`, `IN`, `startsWith`
- `storage:gcp.project.id` - Google Cloud Platform Project ID.
operators: `=`, `IN`, `startsWith`
- `storage:aws.account.id` - Amazon Web Services Account ID.
operators: `=`, `IN`, `startsWith`
- `storage:azure.subscription` - Azure subscription.
operators: `=`, `IN`, `startsWith`
- `storage:azure.resource.group` - Azure resource group.
operators: `=`, `IN`, `startsWith`

Read data based on set of record
attributes supported by 'record-level
permissions on Grail'



Entity - Services

```
1 fetch dt.entity.service
2 | fields id, entity.name, dt.security_context
```

Entity query returns entities
with matching
dt.security_context value(s)

14 records Executed at: 12/2/2024, 14:37:07, Timeframe: 12/1/2024, 14:37:07 - 12/2/2024, 14:37:07 ⓘ

id	entity.name	dt.security_context
SERVICE-C3FB2347EC52E636	aws-easytravel-lnx easyTravel JourneyService	easyTravel AWS Linux, easyTravel AWS
SERVICE-F3562EB1A6006F51	aws-easytravel-lnx - easyTravel-Business	easyTravel AWS Linux, easyTravel AWS
SERVICE-B0B6F60715DCB0D6	aws-easytravel-lnx - easyTravel Customer Frontend	easyTravel AWS Linux, easyTravel AWS
SERVICE-8D582233ADD12F3A	aws-easytravel-lnx - Customer Login	easyTravel AWS Linux, easyTravel AWS
SERVICE-1E17BCE54F9789F2	aws-easytravel-lnx easyTravel AuthenticationService	easyTravel AWS Linux, easyTravel AWS
SERVICE-6F109B7AE2845299	aws-easytravel-lnx - Journey Search Service	easyTravel AWS Linux, easyTravel AWS
SERVICE-72DC04145D4F5FD8	aws-easytravel-lnx - CheckDestination	easyTravel AWS Linux, easyTravel AWS
SERVICE-B6A42181E0B7B48D	aws-easytravel-lnx easyTravel VerificationService	easyTravel AWS Linux, easyTravel AWS
SERVICE-E6C1F26F8B8A8CCA	aws-easytravel-lnx - MongoDB Client Service	easyTravel AWS Linux, easyTravel AWS
SERVICE-C76957A8A73A293C	aws-easytravel-lnx - Requests executed in background threads of easytravel.customer.front..	easyTravel AWS Linux, easyTravel AWS
SERVICE-9DD2CFC955F487AC	aws-easytravel-lnx easyTravel ConfigurationService	easyTravel AWS Linux, easyTravel AWS
SERVICE-62C83A2E5651A225	aws-easytravel-lnx - CreditCardValidation	easyTravel AWS Linux, easyTravel AWS
SERVICE-7980CA47F2ACC8FC	aws-easytravel-lnx easyTravel BookingService	easyTravel AWS Linux, easyTravel AWS
SERVICE-9FFD4D061A82C88C	aws-easytravel-lnx - easyTravel Business Backend	easyTravel AWS Linux, easyTravel AWS





Logs query returns log records
with matching dt.host_group.id
value(s)



Logs

```
1 fetch logs
2 | summarize logs = count(), by: {dt.security_context, dt.host_group.id, dt.source_entity}
3 | sort logs desc
```

9 records Executed at: 12/2/2024, 14:19:00, Timeframe: 12/1/2024, 14:19:00 - 12/2/2024, 14:19:00, Scanned bytes: 2 GB ⓘ

dt.security_context ▾ ▾	dt.host_group.id ▾ ▾	dt.source_entity ▾ ▾	logs ▾ ▾	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-1F08E3AE069C398F	32,378	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-6141FDAE2A84512F	13,911	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-2AC82E1D3DD3BCD7	4,991	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-90E0C324D701A6A7	1,745	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-5D12B96EA912FD13	925	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-06343C12BD54100A	159	
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-FDCC6E8541194024	159	
aws-easytravel-lnx	aws-easytravel-lnx	HOST-AB3EB8C0807B7DDA	71	
aws-easytravel-lnx	aws-easytravel-lnx	HOST-50C6EA577C362DF4	53	



 AWS easyTravel Linux Share

Spans query returns log records with matching dt.host_group.id value(s)

History

 Rerun sections

Spans

```
1 fetch spans, samplingRatio: 10
2 | fieldsAdd service = entityName(dt.entity.service)
3 | summarize spans = count(), by: {dt.security_context, dt.host_group.id, service, dt.entity.service}
4 | sort spans desc
```

12 records

Executed at: 12/2/2024, 14:31:37, Timeframe: 12/1/2024, 14:31:36 - 12/2/2024, 14:31:36, Scanned bytes: 194 MB ⓘ

dt.security_context ▾ ▾	dt.host_group.id ▾ ▾	service ▾ ▾	dt.entity.service ▾ ▾	spans ▾ ▾
null	aws-easytravel-lnx	aws-easytravel-lnx - easyTravel Customer Frontend	SERVICE-B0B6F60715DCB0D6	23,319
null	aws-easytravel-lnx	aws-easytravel-lnx - MongoDB Client Service	SERVICE-E6C1F26F8B8A8CCA	8,387
null	aws-easytravel-lnx	aws-easytravel-lnx easyTravel JourneyService	SERVICE-C3FB2347EC52E636	6,679
null	aws-easytravel-lnx	aws-easytravel-lnx easyTravel ConfigurationService	SERVICE-9DD2CFC955F487AC	2,688
null	aws-easytravel-lnx	aws-easytravel-lnx - CheckDestination	SERVICE-72DC04145D4F5FD8	2,128
null	aws-easytravel-lnx	aws-easytravel-lnx - Requests executed in background threads of easytravel.custo...	SERVICE-C76957A8A73A293C	830
null	aws-easytravel-lnx	aws-easytravel-lnx easyTravel AuthenticationService	SERVICE-1E17BCE54F9789F2	591
null	aws-easytravel-lnx	aws-easytravel-lnx - Journey Search Service	SERVICE-6F109B7AE2845299	551
null	aws-easytravel-lnx	aws-easytravel-lnx - Customer Login	SERVICE-8D582233ADD12F3A	511
null	aws-easytravel-lnx	aws-easytravel-lnx easyTravel BookingService	SERVICE-7980CA47F2ACC8FC	161
null	aws-easytravel-lnx	aws-easytravel-lnx easyTravel VerificationService	SERVICE-B6A42181E0B7B48D	108
null	aws-easytravel-lnx	aws-easytravel-lnx - CreditCardValidation	SERVICE-62C83A2E5651A225	70





Notebooks

+ Notebook

Upload



AWS easyTravel Linux



Share

Metrics query returns log records with matching dt.host_group.id value(s)

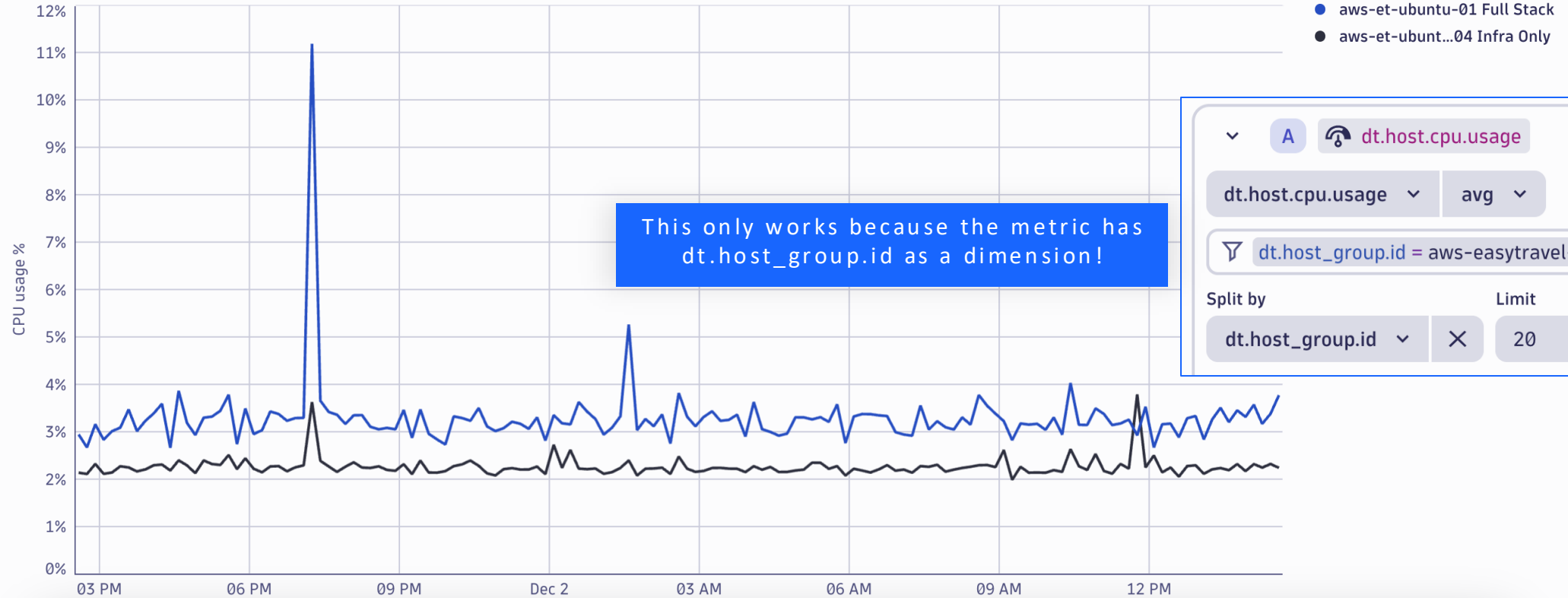
History

Rerun sections

Metrics - Host CPU

```
1 timeseries cpu = avg(dt.host.cpu.usage), by: {dt.entity.host}
2 | fieldsAdd host = entityName(dt.entity.host)
```

2 records Executed at: 12/2/2024, 14:33:55, Timeframe: 12/1/2024, 14:30:00 - 12/2/2024, 14:40:00 ⓘ



This only works because the metric has dt.host_group.id as a dimension!

dt.host.cpu.usage avg

dt.host_group.id = aws-easytravel-lnx

Split by dt.host_group.id Limit 20



Account Management

Home

Subscription

Identity & access management

Lens

Settings

Dynatrace

Groups

View group Ansible Group

View and edit group permissions.

Details

Group name

Ansible Group

Group description

No description

Group type

LOCAL

Edit

Permissions

Permission	Permission Type		Actions
View environment	ROLE	Environment (hfn13693:Ansible)	⋮
Manage monitoring settings	ROLE	Environment (hfn13693:Ansible)	⋮
AppEngine - All Apps	POLICY	Environment (tpc-demo)	⋮
Data Access - Ansible	POLICY	Environment (tpc-demo)	⋮
AppEngine - Notebooks - Restricted User	POLICY	Environment (tpc-demo)	⋮

10 rows per page

Grail permissions based on IAM Policy statements

Policy name*

Data Access - Ansible

Policy description

Read access to data in Grail for Ansible

Policy statement*

1

2

3

4

5

6

7

ALLOW storage:buckets:read;

ALLOW storage:system:read;

ALLOW storage:entities:read WHERE storage:dt.security_context IN ('aws-ansible','Ansible');

ALLOW storage:events:read WHERE storage:dt.host_group.id = 'aws-ansible';

ALLOW storage:logs:read WHERE storage:dt.host_group.id = 'aws-ansible';

ALLOW storage:metrics:read WHERE storage:dt.host_group.id = 'aws-ansible';

ALLOW storage:bizevents:read WHERE

[Learn how to manage policy statements](#)

Read entities based on dt.security_context value

Read data based on dt.host_group.id value

Delete group

Edit

+ Permission



User now has IAM access to
both Ansible and AWS
easyTravel Linux

Entity query returns entities
with matching
dt.security_context value(s)

18 records Executed at: 12/2/2024, 14:45:32, Time taken: 12/1/2024, 14:45:32 - 12/2/2024, 14:45:32 ⓘ

id ⚡ ⌵	entity.name ⚡ ⌵	dt.security_context ⚡ ⌵
SERVICE-C3FB2347EC52E636	aws-easytravel-lnx easyTravel JourneyService	easyTravel AWS Linux, easyTravel AWS
SERVICE-F3562EB1A6006F51	aws-easytravel-lnx - easyTravel-Business	easyTravel AWS Linux, easyTravel AWS
SERVICE-B0B6F60715DCB0D6	aws-easytravel-lnx - easyTravel Customer Frontend	easyTravel AWS Linux, easyTravel AWS
SERVICE-8D582233ADD12F3A	aws-easytravel-lnx - Customer Login	easyTravel AWS Linux, easyTravel AWS
SERVICE-B11B493347B992E1	aws-ansible - host.containers.internal:443	Ansible
SERVICE-1E17BCE54F9789F2	aws-easytravel-lnx easyTravel AuthenticationService	easyTravel AWS Linux, easyTravel AWS
SERVICE-300550AAB9828818	aws-ansible - ip-172-31-38-80.ec2.internal:443	Ansible
SERVICE-6F109B7AE2845299	aws-easytravel-lnx - Journey Search Service	easyTravel AWS Linux, easyTravel AWS
SERVICE-27A2DF5F0E01A6CA	aws-ansible - _:80,443	Ansible
SERVICE-72DC04145D4F5FD8	aws-easytravel-lnx - CheckDestination	easyTravel AWS Linux, easyTravel AWS
SERVICE-B6A42181E0B7B48D	aws-easytravel-lnx easyTravel VerificationService	easyTravel AWS Linux, easyTravel AWS
SERVICE-57CFF2B055896D2A	aws-ansible - uwsgi via domain socket /var/run/tower/uwsgi.sock	Ansible
SERVICE-E6C1F26F8B8A8CCA	aws-easytravel-lnx - MongoDB Client Service	easyTravel AWS Linux, easyTravel AWS
SERVICE-C76957A8A73A293C	aws-easytravel-lnx - Requests executed in background threads of easytravel.customer.front...	easyTravel AWS Linux, easyTravel AWS
SERVICE-9DD2CFC955F487AC	aws-easytravel-lnx easyTravel ConfigurationService	easyTravel AWS Linux, easyTravel AWS

```
1 fetch logs
2 | summarize logs = count(), by: {dt.security_context, dt.host_group.id, dt.source_entity}
3 | sort logs desc
```

22 records Executed at: 12/2/2024, 14:45:32, Timeframe: 12/1/2024, 14:45:32 - 12/2/2024, 14:45:32, Scanned bytes: 2 GB ⓘ

dt.security_context ⚡ ⌵	dt.host_group.id ⚡ ⌵	dt.source_entity ⚡ ⌵	logs ⚡ ⌵
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-1F08E3AE069C398F	32,421
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-6141FDAE2A84512F	13,914
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-D7BE6010DCCA71BE	8,810
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-71D9C5723081A753	6,958
aws-ansible	aws-ansible	PROCE	5,401
aws-easytravel-lnx	aws-easytravel-lnx	PROCE	4,992
aws-ansible	aws-ansible	PROCE	4,202
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-C97CACD04F92349A	3,653
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-69B8E6EAE96D5D9D	2,422
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-90E0C324D701A6A7	1,746
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-B63D4F5D67CA9A86	1,415
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-3807403B1AEC53C7, PROCESS_GROUP_INSTANCE-71D9C57...	1,032
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-5D12B96EA912FD13	910
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-14EEBE83506CE865, PROCESS_GROUP_INSTANCE-D191362...	552
aws-ansible	aws-ansible	PROCESS_GROUP_INSTANCE-14EEBE83506CE865	226
aws-easytravel-lnx	aws-easytravel-lnx	PROCESS_GROUP_INSTANCE-06343C12BD54100A	159

Logs query returns log records with matching dt.host_group.id value(s)

IAM Policy Statements - Segments

- <https://docs.dynatrace.com/docs/manage/identity-access-management/permission-management/manage-user-permissions-policies/advanced/iam-policystatements#storage-filter-segments-read>

Policy name*

Segments - Manage (Admin)

Policy description

Grants access to manage Segments with Admin permission.

Policy statement*

1	ALLOW storage:filter-segments:read;
2	ALLOW storage:filter-segments:write;
3	ALLOW storage:filter-segments:delete;
4	ALLOW storage:filter-segments:admin;

[Learn how to manage policy statements](#)

Policy name*

Segments - Manage (Non-Admin)

Policy description

Grants access to manage Segments without Admin permissions.

Policy statement*

1	ALLOW storage:filter-segments:read;
2	ALLOW storage:filter-segments:write;
3	ALLOW storage:filter-segments:delete;

[Learn how to manage policy statements](#)

Policy name*

Segments - Read

Policy description

Provides read access to use Segments.

Policy statement*

1	ALLOW storage:filter-segments:read;
---	-------------------------------------

[Learn how to manage policy statements](#)



IAM Policy Statements – Default Policies

Effective Policies

Search users to find policy access.

Find user permissions

User*
anthony.pope-cruz@dynatr...
Environments
All
Find Permissions
Clear inputs

The default (out of the box) policies for Admin and User include the policy statements for managing Segments

Results for 'anthony.pope-cruz@dynatrace.com'

Filter
segments
Effect
All
Service
All
Policies
AppEngine - Admi...
Groups
All
reset
Reset filters

Effect	Service	Permission	Has conditions	Policy name	Binding level	Group name
> ALLOW	Storage	filter-segments:admin	No	AppEngine - Admin	Environment	Monitoring admin
> ALLOW	Storage	filter-segments:admin	No	AppEngine - Admin	Environment	Account manager
> ALLOW	Storage	filter-segments:admin	No	AppEngine - Admin	Environment	Monitoring poweruser
> ALLOW	Storage	filter-segments:delete	No	AppEngine - Admin	Environment	Monitoring admin
> ALLOW	Storage	filter-segments:delete	No	AppEngine - Admin	Environment	Account manager
> ALLOW	Storage	filter-segments:delete	No	AppEngine - Admin	Environment	Monitoring poweruser
> ALLOW	Storage	filter-segments:read	No	AppEngine - User	Environment	Dynatrace Peers - Gen3 Power User
> ALLOW	Storage	filter-segments:read	No	AppEngine - User	Environment	Dynatrace Peers
> ALLOW	Storage	filter-segments:read	No	AppEngine - User	Environment	team-wwse-txmv
> ALLOW	Storage	filter-segments:read	No	AppEngine - Admin	Environment	Monitoring admin

10 rows per page





Segments



AWS easyTravel Linux

+ Description

Owner



T Pope-Cruz

Visibility



Unlisted

Variables

Create variables to apply in your Segment data filters

Segment data

Include all data that should be accessible when applying the Segment

+ Variable

Host + Related entity



hostGroupName = "aws-easytravel-lnx"



Run query

Entity filter to include Hosts with relevant hostGroupName and related entities

Service instance runs_on Host



Type to filter

Run query

Process belongs_to Host



Type to filter

Run query

This configuration alone only includes the entities, not the data! (logs, metrics, traces)

Service runs_on Host



Type to filter

Run query

Process Group runs_on Host





+ Description

Owner

T T Pope-Cruz

Visibility

Unlisted

Variables

Create variables to apply in your Segment data filters

+ Variable

Data type filters to include data with matching attributes, such as `dt.security_context` and `dt.host_group.id`

Segment data

Include all data that should be accessible when applying the Segment

 All data types

```
dt.security_context = "aws-easytravel-lnx" OR dt.host_group.id = "aws-easytravel-lnx"
```

▶ Run query

 Preview

50 records



Executed at: December 2, 2024, 3:39:05 PM Timeframe: 1:39:05 PM - 3:39:05 PM

Show results found for

Metrics

Logs

Events

Business events

Spans

timestamp	content	aws.region	aws.resource.id	dt.entity
12/2/2024, 2:17:01 PM	Dec 2 20:17:01 ip-172-31-63-127 CRON[1040374]: pam_unix(cron:session): session ope...	us-east-1	i-0a1127473b5144f54	HOST-A
12/2/2024, 2:17:01 PM	Dec 2 20:17:01 ip-172-31-63-127 CRON[1040374]: pam_unix(cron:session): session clos...	us-east-1	i-0a1127473b5144f54	HOST-A
12/2/2024, 3:17:01 PM	Dec 2 21:17:01 ip-172-31-63-127 CRON[1043704]: pam_unix(cron:session): session ope...	us-east-1	i-0a1127473b5144f54	HOST-A
12/2/2024, 3:17:01 PM	Dec 2 21:17:01 ip-172-31-63-127 CRON[1043704]: pam_unix(cron:session): session clos...	us-east-1	i-0a1127473b5144f54	HOST-A
12/2/2024, 2:17:01 PM	Dec 2 20:17:01 ip-172-31-63-48 CRON[361177]: pam_unix(cron:session): session opene...	us-east-1	i-06487a8b95e6993bc	HOST-50
12/2/2024, 2:17:01 PM	Dec 2 20:17:01 ip-172-31-63-48 CRON[361177]: pam_unix(cron:session): session closed...	us-east-1	i-06487a8b95e6993bc	HOST-50
12/2/2024, 3:17:01 PM	Dec 2 21:17:01 ip-172-31-63-48 CRON[366292]: pam_unix(cron:session): session opene...	us-east-1	i-06487a8b95e6993bc	HOST-50
12/2/2024, 3:17:01 PM	Dec 2 21:17:01 ip-172-31-63-48 CRON[366292]: pam_unix(cron:session): session closed...	us-east-1	i-06487a8b95e6993bc	HOST-50





Segments



< All segments



Ansible

+ Description



Variables

Create variables to apply in your Segment data filters



Segment data

Include all data that should be accessible when applying the Segment



+ Variable

Repeat the process when building the additional Segment for the other application stack

Owner



T Pope-Cruz

Visibility

Unlisted



All data types



dt.security_context = "aws-ansible" OR dt.host_group.id = "aws-ansible"



Run query

Host + Related entity



hostGroupName = "aws-ansible"



Run query

Service instance runs_on Host



Type to filter

Run query

Service runs_on Host



Type to filter

Run query



🏠

Notebooks

+ Notebook

📁 Upload

?

🔍

☰

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Segments - Ansible and AWS easyTravel Linux

⋮

🔄

🔒 Share

🕒 History

🔄 Rerun sections

Spans

⋮

▶ Run

📦 2 segments

🕒 Last 24 hours

🔕 Hide input

⚙ Options

⋮

1 fetch spans, samplingRatio: 10

2 | fieldsAdd service = entityName(dt.

3 | summarize spans = count(), by: {dt

4 | sort spans desc

0 records

Executed at: 12/2/2024, 15:47:06, Timeframe: 12/1/20

Filter by segments

Ansible

×

AWS easyTravel Linux

×

+ Segment

[Manage segments](#)

Clear all

Apply

Applying multiple Segment filters will pre-filter the data (spans) query on records that match BOTH Segment definitions

No data returned!
There are zero records where
dt.host_group.id == "aws-easytravel-
lnx" AND dt.host_group.id == "aws-
ansible"

AND logic, not OR logic!!!

There are no records
Try changing the timeframe or another parameter.

Possible solution?
Dynamic Segment 'Host Group' with
variable for dt.host_group.id and
dt.security_context values!



Kubernetes Namespace

- Application deployed in Kubernetes namespace on shared cluster

k8s-easytravel-aks



Management zone name

k8s-easytravel-aks

Be careful when renaming - if there are policies that are referencing this Management zone, they will need to be adapted to the new name!

Description

Add a description

Rules

Add a new rule

Filter by

Entity selector rules to include all Kubernetes entities related to the namespace

Enabled	Summary	Delete	Details
	Entities matching <code>type("APPLICATION"),fromRelationships.CALLS(type("SERVICE"),toRelationship.isNamespaceOfService(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))</code>		
	Entities matching <code>type("CLOUD_APPLICATION_INSTANCE"),toRelationship.isNamespaceOfCai(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")</code>		
	Entities matching <code>type("CLOUD_APPLICATION"),toRelationship.isNamespaceOfCa(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("CONTAINER_GROUP_INSTANCE"),fromRelationship.isCgiOfNamespace(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("CONTAINER_GROUP"),toRelationship.isPgOfCg(type("PROCESS_GROUP"),toRelationship.isNamespaceOfPg(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))</code>		
	Entities matching <code>type("HTTP_CHECK"),fromRelationship.MONITORS(type("APPLICATION"),fromRelationships.CALLS(type("SERVICE"),toRelationship.isNamespaceOfService(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))))</code>		
	Entities matching <code>type("KUBERNETES_CLUSTER"),fromRelationship.isClusterOfNamespace(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("KUBERNETES_NODE"),toRelationship.isClusterOfNode(type("KUBERNETES_CLUSTER"),fromRelationship.isClusterOfNamespace(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))</code>		
	Entities matching <code>type("KUBERNETES_SERVICE"),toRelationship.isNamespaceOfKubernetesSvc(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("PROCESS_GROUP_INSTANCE"),fromRelationship.isInstanceOf(type("PROCESS_GROUP"),toRelationship.isNamespaceOfPg(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))</code>		
	Entities matching <code>type("PROCESS_GROUP"),toRelationship.isNamespaceOfPg(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks"))</code>		
	Entities matching <code>type("SERVICE"),tag(Tier:Database),toRelationship.CALLS(type("SERVICE"),toRelationship.isNamespaceOfService(type("CLOUD_APPLICATION_NAMESPACE"),entityName.equals("easytravel-aks")))</code>		



View group k8s-easytravel-aks Group

View and edit group permissions.

Details

Edit

Group name
k8s-easytravel-aks Group

Group description
No description

Group type
LOCAL

Permissions

Permission	Permission Type	Scope	Actions
View environment	ROLE	Environment (hfn13693:k8s-easytravel-aks)	
Manage monitoring settings	ROLE	Environment (hfn13693:k8s-easytravel-aks)	

10 rows per page

Page 1 of 1

Permission name

View environment

Permission description

View environment

Permission type

role

Scope

Environments

Select environments

Management zones

k8s-easytravel-aks

Summary

Role is scoped for 1 management zone: hfn13693:k8s-easytravel-aks

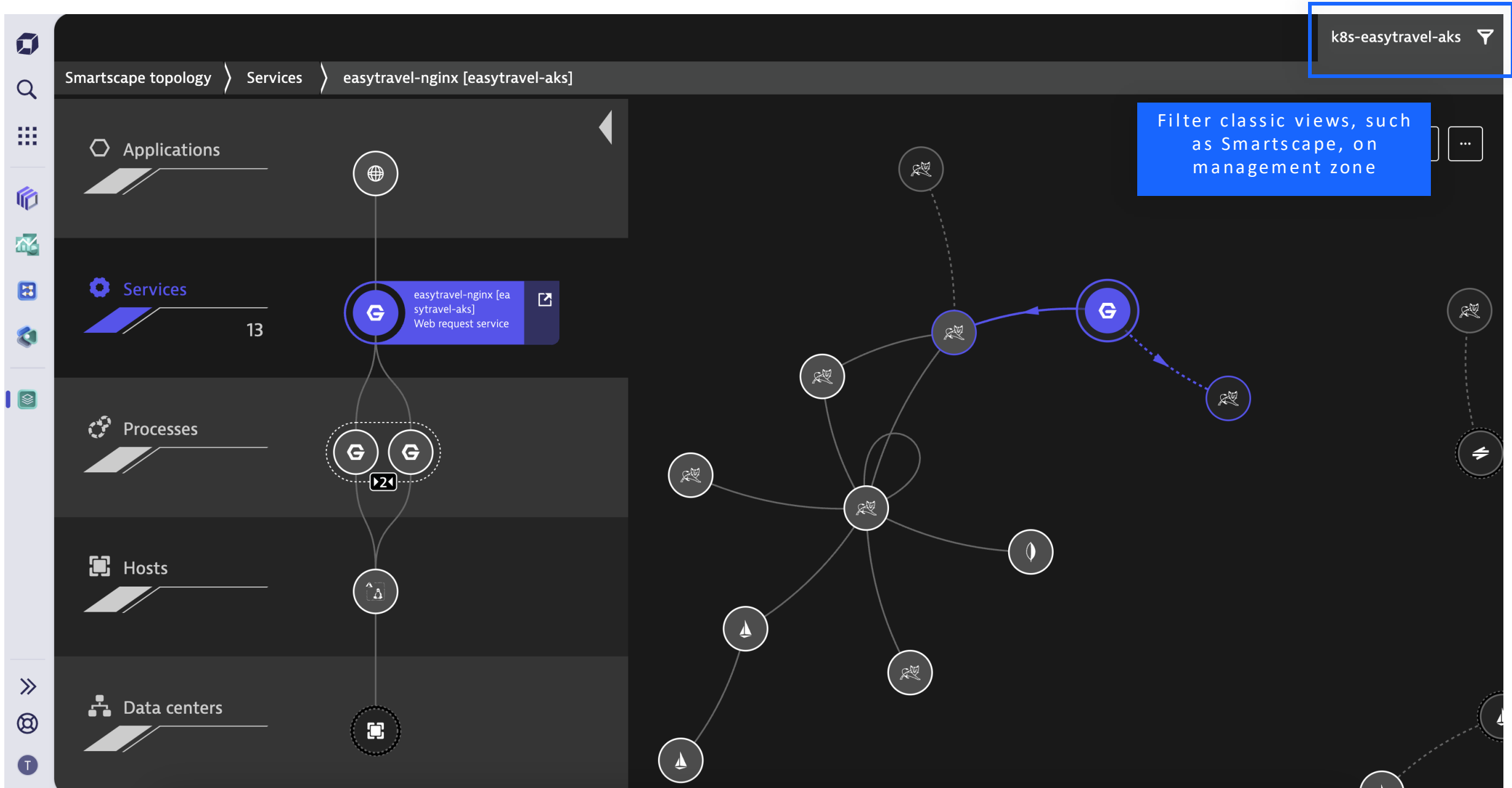
Classic role-based access to only view and/or manage entities/data within management zone

Delete group

Edit

+ Permission





<https://docs.dynatrace.com/docs/manage/identity-access-management/permission-management/manage-user-permissions-policies/iam-policy-boundaries>

Account Management

Home

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Policies

Policies

Boundaries

Policy boundaries

Define policy boundaries that clearly specify whether an action in Dynatrace is allowed.

[Learn how to manage policy boundaries](#)

Boundary

Filter boundaries

Name	Source	Actions
Kubernetes Namespace - easytravel-aks	Custom	

10 rows per page

Page 1 of 1

Policy boundaries make IAM policies modular, separating the permission statement from the where condition statement

Boundary name*

Kubernetes Namespace - easytravel-aks

Boundary query*

1

storage:k8s.namespace.name = "easytravel-aks";

Boundary query acts as WHERE clause condition

+ Boundary

Policy name*

Data Access - Kubernetes Namespace with Boundary

Policy description

Read access to data in Grail for Kubernetes Namespace with dynamic Boundary

Policy statement*

1	ALLOW storage:buckets:read;
2	ALLOW storage:system:read;
3	ALLOW storage:events:read;
4	ALLOW storage:logs:read;
5	ALLOW storage:metrics:read;
6	ALLOW storage:bizevents:read;
7	ALLOW storage:spans:read;

Generic policy granting access to all data types in Grail without conditions

[Learn how to manage policy statements](#)

Permission name

Data Access - Kubernetes Namespace with Boundary

Permission description

Read access to data in Grail for Kubernetes Namespace with dynamic Boundary

Permission type

policy

Policy boundary applies conditions to populate WHERE clause behind the scenes

Scope

☐ Account (all environments)

hfn13693 - tpc-demo

Boundaries

You can view/edit policy boundaries in [Policy management](#)

Kubernetes Namespace - easytravel-aks

Summary

Policy is scoped for 1 environment: hfn13693

Policy is bounded with 1 boundary: Kubernetes Namespace - easytravel-aks



View group k8s-easytravel-aks Group

View and edit group permissions.

Details

Group name
k8s-easytravel-aks Group

Group description
No description

Group type
LOCAL

Edit

Permissions

Permission	Permission Type		Actions
View environment	ROLE	Environment (mm13693:k8s-easytravel-aks)	⋮
Manage monitoring settings	ROLE	Environment (hfn13693:k8s-easytravel-aks)	⋮
Data Access - Kubernetes Namespace easytravel-aks	POLICY	Environment (tpc-demo)	⋮
AppEngine - Notebooks - Restricted User	POLICY	Environment (tpc-demo)	⋮
AppEngine - All Apps	POLICY	Environment (tpc-demo)	⋮
Data Access - Kubernetes Namespace with Boundary	POLICY	Environment (tpc-demo)	⋮

Policy name*

Data Access - Kubernetes Namespace easytravel-aks

Policy description

Read access to data in Grail for Kubernetes Namespace easytravel-aks

Policy statement*

```
1 ALLOW storage:entities:read WHERE
  storage:dt.security_context = 'k8s-easytravel-aks';
```

Entity permissions do not support policy boundaries, until entities move to Grail

[Learn how to manage policy statements](#)

Delete group

Edit

+ Permission





Segments



< All segments



k8s-namespace

Kubernetes Namespace

Use Variables in Segments to create Dynamic Segments

Owner

T T Pope-Cruz

Visibility

Unlisted

Variables

Create variables to apply in your Segment data filters

Segment data

Include all data that should be accessible when applying the Segment

\$name

easytravel-aks

\$id

CLOUD_APPLICATION_NAM...



Variables configuration



Values

```
1 fetch dt.entity.cloud_application_namespace
2 | filterOut in(entity.name, "default", "kube-public", "kube-node-lease", "kube-system", "kube-public")
3 | fields name = entity.name, id
```

Variables based on DQL query

Run query

Open in Notebooks

1 record



Executed at: December 2, 2024, 4:39:33 PM Timeframe: 2:39:34 PM - 4:39:34 PM

Preview

name

id

easytravel-aks

CLOUD_APPLICATION_NAMESPACE-7564A140BA2430F2

The following variables will be created

\$name

easytravel-aks

\$id

CLOUD_APPLICATION_NAM...

Cancel

Done





Segments





k8s-namespace

Kubernetes Namespace

Owner

T T Pope-Cruz

Visibility

Unlisted

Variables

Create variables to apply in your Segment data filters

\$name

easytravel-aks

\$id

CLOUD_APPLICATION_NAM...



Segment data

Include all data that should be accessible when applying the Segment

Kubernetes namespace

+ Related entity

 entity.name = "\$name"

Entity filter in Segment based on selected variable value



 Run query



Container group instance belongs_to Kubernetes namespace

 Type to filter

 Run query



Service belongs_to Kubernetes namespace

 Type to filter

 Run query



Process Group belongs_to Kubernetes namespace

 Type to filter

 Run query



Kubernetes pod belongs_to Kubernetes namespace







CONFIDENTIAL



Include all data that should be accessible when applying the Segment

✓



Data filter in Segment based on selected variable values



Executed with:

```
$name: easytravel-aks
```

```
$id: CLOUD_APPLICATION_NAMESP...
```

At: December 2, 2024, 4:43:34 PM

X

Timeframe: 2:43:34 PM - 4:43:34 PM

Spans



Entity - Services

```
1 fetch dt.entity.service
2 | fields id, entity.name, dt.
```

10 records Executed at: 12/2/2024, 16:44:58, Timefr

id 

Filter by segments

k8s-namespace

easytravel-aks

+ Segment

Manage segments

Apply

Applying the Segment filter will pre-filter the entity query on entities matching the Segment definition

id	el	Apply	
SERVICE-1ADB907C6A750A54	ea		Kubernetes, k8s-easytravel, easyTravel EKS, Azure
SERVICE-3174F290B0B9F855	tpc-aks-demo - easytravel-angular-frontend	k8s-easytravel-aks, Kubernetes, OpenTelemetry	
SERVICE-89069B5ED17725F5	CheckDestinationService [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	
SERVICE-991D984498EC15F9	easytravel-nginx [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	
SERVICE-D712F1BFE46B2007	tpc-aks-demo - easytravel-nginx	k8s-easytravel-aks, Kubernetes, OpenTelemetry	
SERVICE-F0644A49D8AAA318	aks easytravel-aks Requests executed in background t...	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	
SERVICE-78F01A3E67A5DDF5	tpc-aks-demo - easytravel-backend	k8s-easytravel-aks, Kubernetes, OpenTelemetry	
SERVICE-14DF912141204861	easytravel-angular-frontend [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	
SERVICE-91394FE9C528092A	CustomerLogin [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	
SERVICE-D715056066D2E9C2	easytravel-backend [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easyTravel EKS, Azure	



Notebooks + Notebook Upload

Admin - Kubernetes k8s-easytravel-aks Share

History Rerun sections

Run k8s-namespace (3) Last 24 hours Hide input Options

```
1 fetch dt.entity.service
2 | fields id, entity.name, dt.security_context
3 | sort dt.security_context desc
```

48 records Executed at: 12/2/2024, 16:52:30, Timeframe: 12/1/2024, 16:52:30 - 12/2/2024, 16:52:30

id	entity.name	dt.security_context
SERVICE-991D984498EC15F9	easytravel-nginx [easytravel-aks]	easytravel-nginx, K
SERVICE-F0644A49D8AAA318	aks easytravel-aks Requests executed in background threads of tomcat ...	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-14DF912141204861	easytravel-angular-frontend [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-89069B5ED17725F5	CheckDestinationService [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-D715056066D2E9C2	easytravel-backend [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-91394FE9C528092A	CustomerLogin [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-1ADB907C6A750A54	easyTravel-Business DB [easytravel-aks]	k8s-easytravel-aks, Kubernetes, k8s-easytravel, easy
SERVICE-D712F1BFE46B2007	tpc-aks-demo - easytravel-nginx	k8s-easytravel-aks, Kubernetes, OpenTelemetry
SERVICE-78F01A3E67A5DDF5	tpc-aks-demo - easytravel-backend	k8s-easytravel-aks, Kubernetes, OpenTelemetry
SERVICE-3174F290B0B9F855	tpc-aks-demo - easytravel-angular-frontend	k8s-easytravel-aks, Kubernetes, OpenTelemetry
SERVICE-A9885C924BEC893C	eks easytravel-prod Java Frontend	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-AD085A6CA5A0595F	eks easytravel-prod AuthenticationService	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-DCF70038864DB041	eks easytravel-prod easyTravel Business Backend	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-5AAC3159E253E5A2	eks easytravel-prod EasytravelService	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-F7F61BC150373D1C	eks easytravel-prod BookingService	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-719FDD47599045E6	eks easytravel-prod tomcat easytravel-frontend-* on port 8080	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,
SERVICE-EE35311C7179AEB8	eks easytravel-prod DB	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS,

Filter by segments

k8s-namespace easytravel-nonprod, easyt... X

+ Segment Search

- ☒ easytravel-nonprod
- ☒ easytravel-prod
- ☒ easytravel-aks
- ☐ easytrade-nonprod
- ☐ argocd
- ☐ dynatrace
- ☐ prometheus
- ☐ opentelemetry-operator-system
- ☐ app-routing-system
- ☐ istio-system
- ☐ astronomy-shop
- ☐ springio
- ☐ cert-manager
- ☐ easytrade-prod

Applying multiple values of Dynamic Segment filter uses OR logic, includes any entity (or data) that matches at least value selected from list

Admin - Kubernetes k8s-easytravel-aks Share

2 segments Last 24 hours Hide input Options

```
1 fetch dt.entity.service
2 | fields id, entity.name, dt.security_context
3 | sort dt.security_context desc
```

19 records Executed at: 12/2/2024, 16:54:32, Timeframe: 12/1/2024, 16:54:31 - 12/2/2024, 16:54:31

id	entity.name	dt.security_context
SERVICE-A9885C924BEC893C	eks easytravel-prod Java Frontend	AWS, [Kubernetes] tpc-de
SERVICE-AD085A6CA5A0595F	eks easytravel-prod AuthenticationService	AWS, [Kubernetes] tpc-de
SERVICE-DCF70038864DB041	eks easytravel-prod easyTravel Business Backend	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS, Ansible Self-Healing, Kubernetes, k8s-easytravel-prod, k8s-easytravel, ...
SERVICE-5AAC3159E253E5A2	eks easytravel-prod EasytravelService	AWS, [Kubernetes] tpc-demo-2-eks-prod, AWS EKS, Ansible Self-Healing, Kubernetes, k8s-easytravel-prod, k8s-easytravel, ...

Filter by segments

k8s-namespace easytravel-nonprod, easyt... X

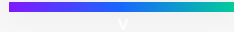
owner easytrade-devops, easytr... X

+ Segment Manage segments Clear all Apply

Adding multiple Dynamic Segment filters uses AND logic: namespace = "a" or "b" or "c" AND owner = "x" or "y" or "z"



Hands On



v

Hands On

- Environment Set Up
 - Dynatrace Setup
 - GitHub Repository & Codespaces Setup
 - IDP Setup & Validation
- Self-Service Onboarding
 - Onboard New Apps
 - Explore New Apps
 - Analyze without Segments
 - Kubernetes App, Dashboards App, Notebooks App
- Basic Segments
 - Kubernetes
 - Cluster, Namespace
 - Analyze with Segments
- OpenPipeline
 - Logs Pipeline
 - Metrics Pipeline
- Complex Segments
 - IDP Project Team
 - Analyze with Segments



Wrap Up

Key Takeaways

- Management Zones are still supported for Gen2 use cases
 - Customers can continue to manage access and filter visualizations to Classic Dynatrace using MZ
- Managing access and filtering visualizations with Grail uses IAM + Segments
 - Manage access to data in Grail with IAM Policies, including record-level permission statements
 - Filter visualizations with Grail using Segments
- Customers should strategize their IAM Policies for data access management first before developing Segment configurations
 - Segments are quick and reusable DQL query filters, not like management zones!
- Segment feature improvements are coming
 - Many of today's limitations are due to the entity model not being in Grail
- Best practice implementation sequence:
 - Ingest data into Dynatrace, Configure OpenPipeline, Configure IAM, Configure Segments



CLOUD DONE RIGHT